Properties of

1) $\vec{U}\cdot\vec{V} = \vec{V}\cdot\vec{U}$
2) $\vec{U}\cdot(\vec{V}+\vec{W}) = \vec{U}\cdot\vec{V} + \vec{U}\cdot\vec{W}$
3) $(c\vec{U})\cdot\vec{V} = c(\vec{U}\cdot\vec{V})$
4) $\vec{U}\cdot\vec{U} \geq 0$ $\quad \vec{U}\cdot\vec{U} = 0$ iff $\vec{U} = 0$

Define eigen values.

Eigen values:

over R

⇒ if A is $n \times n$ matrix, then λ is an eigen value of A if and only if it satisfies the equation

$$\det(AI - \lambda I) = 0 \quad \text{(singular)}$$

⇒ This is called characteristic equation of A.

In other words:

"Let A be an $n \times n$ matrix over R, the number λ is an eigen value of A if there exist a non-zero vector in $\mathbb{R}^n$ $v \in$ such that

$$A\vec{x} = \lambda\vec{x}$$

Here $\vec{x}$ is called eigen vector of "A" associated with eigen value λ."

$\Rightarrow$ λ is an eigen value if $A\vec{x} = \lambda\vec{x}$ has a non-trivial solution.

Eigen vector:

$\Rightarrow$ An eigen vector of an $n \times n$ matrix A is a vector $\vec{x}$ such that

$$A\vec{x} = \lambda\vec{x}$$

for some scalar λ and $\vec{x} \neq 0$.

Characteristic equation:

- Let A be $n \times n$ matrix, then λ is an eigen value of A if and only if it satisfies the equation

$$\det(A - I\lambda) = 0 \quad \text{(singular)}$$

- This is called characteristic equation of A.

for 2x2 matrix

$$\lambda^2 - (\text{Trace of } A)\lambda + \det A = 0$$

for 3x3 matrix

$$\lambda^3 - (\text{Trace of } A)\lambda^2 + \left(\text{Sum of minors along diagonal}\right)\lambda - \det A = 0$$

Sum of minors along diagonal $= M_{11} + M_{22} + M_{33}$? find

Question:

$$A = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$$

Let $A = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$

$\therefore$ Let λ be the eigen value of A so $|A - I\lambda| = 0$

$$\begin{vmatrix} 4-\lambda & 0 & 1 \\ -2 & 1-\lambda & 0 \\ -2 & 0 & 1-\lambda \end{vmatrix} = 0.$$

$$(4-\lambda)\begin{vmatrix} 1-\lambda & 0 \\ 0 & 1-\lambda \end{vmatrix} - 0\begin{vmatrix} -2 & 0 \\ -2 & 1-\lambda \end{vmatrix} + 1\begin{vmatrix} -2 & 1-\lambda \\ -2 & 0 \end{vmatrix} = 0$$

$(4-\lambda)(1-\lambda)^2 - 0 + (0 + 2(1-\lambda)) = 0$

$(4-\lambda)(1-2\lambda+\lambda^2) + 2(1-\lambda) = 0$

$4 - 8\lambda + 4\lambda^2 - \lambda + 2\lambda^2 - \lambda^3 + 2 - 2\lambda = 0$

$-\lambda^3 + 6\lambda^2 - 11\lambda + 6 = 0$

$\lambda = 1$ by synthetic division

$$\begin{array}{c|cccc} & -1 & 6 & -11 & 6 \\ 1 & \downarrow & -1 & 5 & -6 \\ \hline & -1 & 5 & -6 & 0 \end{array}$$

$-\lambda^2+5\lambda-6=0$

$\lambda^2-5\lambda+6=0$

$\lambda^2-3\lambda-2\lambda+6=0$

$\lambda(\lambda-3)-2(\lambda-3)=0$

$(\lambda-2)(\lambda-3)=0$

$\lambda=2\ ,\quad \lambda=3$

$\{1,2,3\}$ are eigen values of matrix A.

Q#2: $A=\begin{bmatrix}1/2 & 0 & 0\\ -1 & 2/3 & 0\\ 5 & -8 & -1/4\end{bmatrix}$

Let λ be the eigen value of A

so, $|A-I\lambda|=0$

$$\begin{bmatrix}1/2 & 0 & 0\\ -1 & 2/3 & 0\\ 5 & -8 & -1/4\end{bmatrix}-\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}\lambda$$

$$\begin{bmatrix}1/2 & 0 & 0\\ -1 & 2/3 & 0\\ 5 & -8 & -1/4\end{bmatrix}-\begin{bmatrix}\lambda & 0 & 0\\ 0 & \lambda & 0\\ 0 & 0 & \lambda\end{bmatrix}$$

$$\begin{bmatrix}\frac{1}{2}-\lambda & 0 & 0\\ -1 & 2/3-\lambda & 0\\ 5 & -8 & -1/4-\lambda\end{bmatrix}$$

$$\begin{vmatrix} \frac{1}{2}-\lambda & 0 & 0 \\ -1 & 2/3-\lambda & 0 \\ 5 & -8 & \frac{-1}{4}-\lambda \end{vmatrix} = 0$$

$$\left(\frac{1}{2}-\lambda\right)\begin{vmatrix} 2/3-\lambda & 0 \\ -8 & -1/4-\lambda \end{vmatrix} = 0$$

$$\left(\frac{1}{2}-\lambda\right)\left[\left(\frac{2}{3}-\lambda\right)\left(\frac{-1}{4}-\lambda\right)-0\right] = 0$$

$\frac{1}{2}-\lambda=0$

$\frac{1}{2}=\lambda$

$\boxed{\lambda=\frac{1}{2}}$

$\left(\frac{2}{3}-\lambda\right)\left(\frac{-1}{4}-\lambda\right)=0$

$\frac{-2}{12}-\frac{2}{3}\lambda+\lambda\left(\frac{1}{4}\right)+\lambda^2=0$

$\lambda^2+\frac{\lambda}{4}-\frac{2\lambda}{3}-\frac{2}{12}=0$

$12\lambda^2+3\lambda-8\lambda-2=0$

$12\lambda^2-5\lambda-2=0$

$12\lambda^2-8\lambda+3\lambda-2=0$

$4\lambda(3\lambda-2)+1(3\lambda-2)=0$

$(4\lambda+1)(3\lambda-2)=0$

$4\lambda+1=0$

$\boxed{\lambda=-1/4}$, $\boxed{\lambda=2/3}$

$\frac{1}{2}, \frac{-1}{4}, \frac{2}{3}$ are eigen values of A.

Q#: Find eigen values and eigen vector of the following matrix

$$A = \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix}$$

Let λ be the eigen values of the following matrix A, so

$$|A - \lambda I| = 0$$

$$\left| \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right| = 0$$

$$\left| \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} \right| = 0$$

$$\begin{vmatrix} 1-\lambda & -3 & 3 \\ 3 & -5-\lambda & 3 \\ 6 & -6 & 4-\lambda \end{vmatrix} = 0$$

$$=(1-\lambda)\begin{vmatrix} -5-\lambda & 3 \\ -6 & 4-\lambda \end{vmatrix} + 3\begin{vmatrix} 3 & 3 \\ 6 & 4-\lambda \end{vmatrix} + 3\begin{vmatrix} 3 & -5-\lambda \\ 6 & -6 \end{vmatrix}$$

$= (1-\lambda)[(-5-\lambda)(4-\lambda)+18] + 3[12-3\lambda-18] +$
$3[-18+30+6\lambda] = 0$

$= (1-\lambda)(-20+5\lambda-4\lambda+\lambda^2+18] + 3[-3\lambda-6] +$
$3[12+6\lambda] = 0$

$= (1-\lambda)(\lambda^2+\lambda-2) + (3)(-3)(\lambda+2) + (3)(6)(2+\lambda)$
$= 0$

$= (1-\lambda)(\lambda^2+2\lambda-\lambda-2) - 9(\lambda+2) + 18(\lambda+2) = 0$
$= (1-\lambda)(\lambda+2)(\lambda-1) - 9(\lambda+2) + 18(\lambda+2) = 0$
$= (\lambda+2)[(1-\lambda)(\lambda-1) - 9 + 18] = 0$
$= (\lambda+2)[\lambda-1-\lambda^2+\lambda+9] = 0$
$= (\lambda+2)(-\lambda^2+2\lambda+8) = 0$
$= (\lambda+2)(-\lambda^2+4\lambda-2\lambda+8) = 0$
$= (\lambda+2)(\lambda-4)(-2-\lambda) = 0$

$\lambda+2=0$	$\lambda-4=0$	$-2-\lambda=0$
$\boxed{\lambda=-2}$	$\boxed{\lambda=4}$	$\boxed{\lambda=-2}$

for $\lambda_1=-2$

Let $V_1 = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ be the eigen vector corresponding to $\lambda_1 = 2$

$|A-(2)I| v_1 = 0$

$$\begin{vmatrix} 1-(-2) & -3 & 3 \\ 3 & -5-(-2) & 3 \\ 6 & -6 & 4-(-2) \end{vmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 3 & -3 & 3 \\ 3 & -3 & 3 \\ 6 & -6 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Now,

$$\begin{bmatrix} 3 & -3 & 3 \\ 3 & -3 & 3 \\ 6 & -6 & 6 \end{bmatrix}$$

$$\sim R_3 \to \frac{1}{2} R_3 \begin{bmatrix} 3 & -3 & 3 \\ 3 & -3 & 3 \\ 3 & -3 & 3 \end{bmatrix}$$

$\sim R_3 \to R_3 - R_2$

$\sim R_2 \to R_2 - R_1$

$$\begin{bmatrix} 3 & -3 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\sim R_1 \to \frac{1}{3} R_1 \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x_1 - x_2 + x_3 = 0$$

Let $x_1 = t$, $x_2 = 2t$

then

$x_3 = -t + 2t$

$x_3 = t$

$$v_1 = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} t \\ 2t \\ -t \end{bmatrix} = t \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

find also v_2

Q#: Find eigen values and eigen vector

$$A = \begin{bmatrix} 5 & 8 & 16 \\ 4 & 1 & 8 \\ -4 & -4 & -11 \end{bmatrix}$$

Let λ be the eigen values of A so, $|A - \lambda I| = 0$

$$\left| \begin{bmatrix} 5 & 8 & 16 \\ 4 & 1 & 8 \\ -4 & -4 & -11 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right| = 0$$

$$\left| \begin{bmatrix} 5 & 8 & 16 \\ 4 & 1 & 8 \\ -4 & -4 & -11 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} \right| = 0$$

$$\begin{vmatrix} 5-\lambda & 8 & 16 \\ 4 & 1-\lambda & 8 \\ -4 & -4 & -11-\lambda \end{vmatrix} = 0$$

$$= (5-\lambda)\begin{vmatrix} 1-\lambda & 8 \\ -4 & -11-\lambda \end{vmatrix} - 8\begin{vmatrix} 4 & 8 \\ -4 & -11-\lambda \end{vmatrix} + 16\begin{vmatrix} 4 & 1- \\ -4 & - \end{vmatrix} = 0$$

$$= (5-\lambda)\left[(1-\lambda)(-11-\lambda)+32\right] - 8\left(4(-11-\lambda)+32\right) + 16(-16+4-4\lambda) = 0$$

$$= (5-\lambda)\left[-11-\lambda+11\lambda+\lambda^2+32\right] - 8\left[-44-4\lambda+32\right] + 16(-12-4\lambda) = 0$$

$$\Rightarrow (5-\lambda)\left[\lambda^2+10\lambda+21\right] - 8\left[-12-4\lambda\right] + 16(-12-4\lambda)$$

$$\Rightarrow (5-\lambda)\left[\lambda^2+3\lambda+7\lambda+21\right] - (8)(-4)(3+\lambda) + 16(-4)(3+\lambda)$$

$$\Rightarrow (5-\lambda)(\lambda+3)(\lambda+7) + 32(\lambda+3) + (-64)(\lambda+3) = 0$$

$$\Rightarrow (5-\lambda)(\lambda+3)(\lambda+7) + 32(\lambda+3) - 64(\lambda+3) = 0$$

$$\Rightarrow (\lambda+3)\left[(5-\lambda)(\lambda+7) + 32 - 64\right] = 0$$

$$(\lambda+3)\left[5\lambda+35-\lambda^2-7\lambda-32\right] = 0$$

$$= (\lambda+3)(-\lambda^2-2\lambda+3) = 0$$

$$(\lambda+3)(-\lambda^2-3\lambda+\lambda+3) = 0$$

$$= (\lambda+3)\left(-\lambda(\lambda+3)+1(\lambda+3)\right) = 0$$

$$(\lambda+3)(1-\lambda)(\lambda+3) = 0$$

$\lambda+3=0$	$1-\lambda=0$
$\boxed{\lambda=-3}$	$\boxed{\lambda=1}$

for $\lambda_1 = -3$

Let $V_1 = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ be the eigen vector

corresponding to $\lambda_1 = -3$

$|A - (-3)I| V_1 = 0$

$$\begin{bmatrix} 8 & 8 & 16 \\ 4 & 4 & 8 \\ -4 & -4 & -8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Now,

$$\begin{bmatrix} 8 & 8 & 16 \\ 4 & 4 & 8 \\ -4 & -4 & -8 \end{bmatrix}$$

$\sim R_3 \to R_3 + R_2$

$\sim \frac{1}{4} R_2$

$\sim \frac{1}{8} R_1$

$$\begin{bmatrix} 1 & 1 & 2 \\ 1 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$\sim R_2 \to R_2 - R_1$

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$x_1 + x_2 + 2x_3 = 0$

let $x_2 = t$

$x_3 = 2t$

$x_1 + t + 4t = 0$

$x_1 + 5t = 0$

$x_1 = -5t$

$$v_1 = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -5t \\ t \\ 2t \end{bmatrix} = t \begin{bmatrix} -5 \\ 1 \\ 2 \end{bmatrix}$$

for $\lambda_2 = 1$

Let $V_2 = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$ be the eigen ve

against $\lambda_2 = 1$

$|A - \lambda I| V_2 = 0$

$$\begin{bmatrix} 4 & 8 & 16 \\ 4 & 0 & 8 \\ -4 & -4 & -\cancel{12}\,0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$\sim R_3 \to -\frac{1}{4} R_3$
$\sim R_2 \to \frac{1}{4} R_2$
$\sim R_1 \to \frac{1}{4} R_1$

$$\begin{bmatrix} 1 & 2 & 8 \\ 1 & 0 & 2 \\ 1 & 1 & 3 \end{bmatrix}$$

$\sim R_3 \to R_3 - R_2$

$$\begin{bmatrix} 1 & 2 & 8 \\ 1 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix}$$

$\sim R_1 \to R_1 - R_2$

$$\begin{bmatrix} 0 & 2 & 6 \\ 1 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix}$$

$\sim R_1 \to R_1 - R_3$

$$\begin{bmatrix} 0 & 1 & 5 \\ 1 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix}$$

$\sim R_3 \to R_3 - R_1$

$$\begin{bmatrix} 0 & 1 & 5 \\ 1 & 0 & 2 \\ 0 & 0 & -4 \end{bmatrix}$$

$-\frac{1}{4} R_3 \to R_3$

$$\begin{bmatrix} 0 & 1 & 5 \\ 1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

$\sim R_2 \to R_2 - R_3$

$$\begin{bmatrix} 0 & 1 & 5 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$y_2 + 5y_3 = 0$ —— (i)

$y_1 + y_3 = 0$ —— (ii)

$\boxed{y_3 = 0}$ —— (iii)

put eq (iii) in eq(i)

$y_1 + 0 = 0$

$\boxed{y_1 = 0}$

Put ~~y=0~~ and $y_3 = 0$ in (i)

$y_2 + 5(0) = 0$

$\boxed{y_2 = 0}$

$$V_1 = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Q# Find Characteristic equation, eigen values and eigen vectors.

(PP-2014)

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 3 & 0 \\ 3 & 2 & -2 \end{bmatrix}$$

Let λ be the eigen value against

$A \Rightarrow 0 \quad |A - \lambda I| = 0$

$$\left| \begin{bmatrix} 1 & 0 & 0 \\ -1 & 3 & 0 \\ 3 & 2 & -2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right| = 0$$

$$\begin{vmatrix} 1-\lambda & 0 & 0 \\ -1 & 3-\lambda & 0 \\ 3 & 2 & -2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda) \begin{vmatrix} 3-\lambda & 0 \\ 2 & -2-\lambda \end{vmatrix} - 0 \begin{vmatrix} -1 & 0 \\ 3 & -2-\lambda \end{vmatrix} + 0 \begin{vmatrix} -1 & 3-\lambda \\ 3 & 2 \end{vmatrix} = 0$$

$$(1-\lambda)(-6-3\lambda+2\lambda+\lambda^2) - 0 + 0 = 0$$

$$\Rightarrow (1-\lambda)(\lambda^2-3\lambda+2\lambda+6) = 0$$

$$\Rightarrow (1-\lambda)(\lambda-3)(\lambda+2) = 0$$

Characteristic equation:

$\Rightarrow (1-\lambda)(\lambda^2-\lambda-6)=0$

$\Rightarrow \lambda^2-\lambda-6-\lambda^3+\lambda^2-6\lambda=0$

$\Rightarrow -\lambda^3+2\lambda^2-7\lambda-6=0$

$\Rightarrow -(\lambda^3-2\lambda^2+7\lambda+6)=0$

$\Rightarrow \boxed{\lambda^3-2\lambda^2+7\lambda+6=0}$

Eigen values:

$1-\lambda=0$	$\lambda-3=0$	$\lambda+2=0$
$\boxed{\lambda_1=1}$	$\boxed{\lambda_2=3}$	$\boxed{\lambda_3=-2}$

Eigen vectors:

for $\lambda_1=1$

Let $V_1=\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ be the eigen vector

against $\lambda_1=1$

$$|A-(1)I|\,V_1=0$$

$$\begin{bmatrix} 0 & 0 & 0 \\ -1 & 2 & 0 \\ 3 & 2 & -3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$R_3 \leftrightarrow R_1 \begin{bmatrix} 3 & 2 & -3 \\ -1 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\sim R_2 \to R_2 + 3R_1 \begin{bmatrix} 3 & 2 & -3 \\ 0 & -4 & -9 \\ 0 & 0 & 0 \end{bmatrix}$$

$-x_1 + 2x_2 = 0$ ———— (i)

$3x_1 + 2x_2 - 3x_3 = 0$ ———— (ii)

$\cancel{-}x_1 = \cancel{-}2x_2$

$x_1 = 2x_2$ ——— (iii)

Let $x_2 = t$

$x_1 = 2t$

put $x_1 = 2t$ & $x_2 = t$ in (ii)

$6t + 2t - 3x_3 = 0$

$8t - 3x_3 = 0$

$\cancel{-}3x_3 = \cancel{-}8t$

$x_3 = 8/3\, t$

$$v_1 = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2t \\ t \\ 8/3\, t \end{bmatrix} = t \begin{bmatrix} 2 \\ 1 \\ 8/3 \end{bmatrix}$$

for $\lambda_2 = 3$
Let $X_2 = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$ be the eigen vector

against $\lambda_2 = 3$

$$|A - 3I| V_2 = 0$$

$$\begin{bmatrix} -2 & 0 & 0 \\ -1 & -5 & 0 \\ 3 & 2 & -5 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Now,

$\sim R_1 \to -\frac{1}{2} R_1$ $\begin{bmatrix} 1 & 0 & 0 \\ -1 & -5 & 0 \\ 3 & 2 & -5 \end{bmatrix}$

$\sim R_3 \to R_3 - 3R_1$ $\begin{bmatrix} 1 & 0 & 0 \\ -1 & -5 & 0 \\ 0 & -1 & -5 \end{bmatrix}$

$\sim R_2 \to R_2 + R_1$ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & -5 & 0 \\ 0 & -2 & -5 \end{bmatrix}$

$\sim R_2 \to -\frac{1}{5} R_2$ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & -5 \end{bmatrix}$

$\sim R_3 \to R_3 + R_2$ $\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 2 & -5 \end{bmatrix}$

$\sim R_3 \to \frac{-1}{5} R_3$ $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 2 & -5 \end{bmatrix}$

$y_1 = 0$?

$y_2 = 0$

$y_3 = 0$

$V_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

for $\lambda_3 = -2$

let $V_3 = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$ be the eigen vector corresponding to $\lambda_3 = -2$

$|A - (-2)I| V_3 = 0$

$\begin{bmatrix} 3 & 0 & 0 \\ -1 & 5 & 0 \\ 3 & 2 & 0 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$R_1 \to \frac{1}{3} R_1$
$R_3 \to R_3 - 3R_1$
$$\begin{bmatrix} 1 & 0 & 0 \\ -1 & 5 & 0 \\ 0 & 2 & 0 \end{bmatrix}$$

$$\sim R_2 \to R_2 + R_1 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 2 & 0 \end{bmatrix}$$

$$\sim \frac{1}{5} R_2 \to R_2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 0 \end{bmatrix}$$

$$\sim R_3 \to R_3 - 2R_2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$Z_1 = 0$

$Z_2 = 0$

$$V_3 = \begin{bmatrix} Z_1 \\ Z_2 \\ Z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \ \end{bmatrix}$$

Write down properties of eigen values.

Properties of eigen values:

⇒ if two eigen values are same and there is only one linearly independent solution then there will be same eigen vector for both the eigen values.

⇒ if two eigen values are same but there is two linearly independent solution then there will be different eigen vector for each eigen values.

⇒ Roots of a characteristics equation are known as eigen values, eigen roots or characteristic roots or latent roots.

⇒ Set consisting eigen vectors corresponding to different eigen values is linearly independent.

⇒ The set of eigen values of the matrix A is called the **spectrum** of the matrix A and the largest eigen value is called the spectral radius of A.

⇒ algebraic multiplicity:

⇒ Number of times a particular eigen value repeats is known as its algebraic multiplicity.

example:

$\lambda = 1, 1, 3$

then multiplicity of 1 is 2 and multiplicity of 3 is 1.

⇒ Geometric multiplicity:

⇒ Number of distinct linear independent eigen vectors for a particular eigen value is known as its geometric multiplicity

$m(1) =$ Nullity $(A - I)$ for example for $\lambda = 7$
$GM(7) = N = |A - 7I|$

Properties:

- Eigen values of a hermitian matrix are all real.
- Eigen values of skew hermitian matrix are either 0 or purely imaginary
- Eigen values of orthogonal matrix are of unit modulus.

- Eigen values of an idempotent matrix are either zero or one.
- Eigen values of a unitary matrix are of unit modulus.
- Eigen values of an involutory matrix are either +1, and -1.
- if A and B are two matrices of same order then A and B have same eigen values where $B = P^{-1}AP$ or similar matrices have same eigen values.

- if λ is an eigen values of A then,
  - λ^{-1} is an eigen value of A^{-1}.
  - if $\lambda_1, \lambda_2, \ldots, \lambda_n$ eigen values of A then eigen values of A' are $\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \ldots, \frac{1}{\lambda_n}$
- Eigen value of adj $A = \left[\frac{|A|}{\lambda}\right]$

A and A^t have same eigen values

if $\lambda_1, \lambda_2, \ldots, \lambda_3$ are then eigen values of square matrix A, then find eigen values and eigen spaces! (PP - 2021)

$$A = \begin{bmatrix} -2 & -1 \\ 5 & 2 \end{bmatrix}$$

$$\Rightarrow \begin{vmatrix} -2-\lambda & -1 \\ 5 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (-2-\lambda)(2-\lambda)+5=0$$

$$-4+2\lambda-2\lambda+\lambda^2+5=0$$

$$\Rightarrow \lambda^2+1=0$$

$$\Rightarrow \lambda^2=-1$$

$\boxed{\lambda = \pm i}$ eigen values

for $\lambda_1 = i$

Let $v_1 = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ be the eigen vector against $\lambda_1 = i$

$$\begin{bmatrix} -2-i & -1 \\ 5 & 2-i \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$5x_1+(2-i)x_2=0$ —— (i)

$(-2-i)x_1 - x_2 = 0$ —— (ii)

$x_1(-2-i) = x_2$

$$\frac{x_1}{1} = \frac{x_2}{(-2-i)}$$

$x_1 = 1$

$x_2 = -2-i$

$$v_1 = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1+0i \\ -2-i \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix} i$$

from eq (i)

$$5x_1 = -(2-i)x_2$$

$$\frac{x_1}{-(2-i)} = \frac{x_2}{5}$$

$$v_1 = \begin{bmatrix} -2+i \\ 5+0i \end{bmatrix} = \begin{bmatrix} -2 \\ 5 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} i$$

for $\lambda_2 = -i$

Let $v_2 = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$ be the eigen vector corresponding to $\lambda_2 = -i$

$$\begin{bmatrix} -2+i & -1 \\ 5 & 2+i \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$(-2+i)y_1 - y_2 = 0 \quad \text{--- (i)}$$

$$5y_1 + (2+i)y_2 = 0 \quad \text{--- (ii)}$$

from (i)

$$y_1(-2+i) = y_2$$

$$\frac{y_1}{1} = \frac{y_2}{(-2+i)}$$

$$v_2 = \begin{bmatrix} 1+0i \\ -2+i \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} i$$

from eq (ii)

$$5y_1 = -(-2+i)y_2$$

$$\frac{y_1}{2-i} = \frac{y_2}{5}$$

$$v_2 = \begin{bmatrix} 2-i \\ 5+0i \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix} + \begin{bmatrix} -1 \\ 0 \end{bmatrix} i$$

Q# find eigen values of

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 4 & -17 & 8 \end{bmatrix} \quad \text{(PP-2020)}$$

$$R_3 \longleftrightarrow R_2 \quad \begin{bmatrix} 0 & 1 & 0 \\ 4 & -17 & 8 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_1 \longleftrightarrow R_2 \quad \begin{bmatrix} 4 & -17 & 8 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$= \begin{vmatrix} 4-\lambda & -17 & 8 \\ 0 & 1-\lambda & 0 \\ 0 & 0 & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (4-\lambda) \begin{vmatrix} 1-\lambda & 0 \\ 0 & 1-\lambda \end{vmatrix} + -17 \begin{vmatrix} 0 & 0 \\ 0 & 1-\lambda \end{vmatrix} + 8 \begin{vmatrix} 0 & 1-\lambda \\ 0 & 0 \end{vmatrix} = 0$$

$\Rightarrow (4-\lambda)(1-\lambda)^2 + 0 + 0 = 0$

$(4-\lambda)(1-\lambda)^2 = 0$

$(1-\lambda)^2 = 0$	$4-\lambda = 0$
$\lambda^2 - 2\lambda + 1 = 0$	$\lambda = 4$
$(\lambda^2 - \lambda - \lambda + 1) = 0$	
$\lambda(\lambda-1) - 1(\lambda-1) = 0$	
$(\lambda-1)(\lambda-1) = 0$	

$\lambda = 1, \lambda = 1$

$\{1, 1, 4\}$ are eigen values.